

BANK LOAN ANALYTICS

End-to-End Business Intelligence Solution

Project Documentation

Prepared using: SQL Server (SSMS) · Power Query · DAX · Power BI

INEX Bank

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Executive Summary

This project delivers a three-page Power BI report — Summary, Overview, and Details — built on top of a single loan-application table (Bank_Loan_data) and validated independently against SQL Server queries. As of the reporting period (December 2021), the portfolio consists of 38,576 loan applications totalling \$435.8M in funded amount and \$473.1M in amount received, at an average interest rate of 12.05% and average DTI of 13.3%.

Portfolio quality is strong: 86.2% of loans are classified as Good (Fully Paid or Current) versus 13.8% Bad (Charged Off). Good loans return more cash than was disbursed, while Charged Off loans recover only around 57% of what was funded — the clearest single risk signal in the report.

Debt consolidation is the dominant loan purpose, mortgage and rental borrowers account for almost all funded amount, and lending volume grew steadily month-over-month through the year. The sections that follow document the objective, data, validation process, data model, and each dashboard page in detail, followed by business insights, project challenges, and recommended future enhancements.

Part 1 — Introduction

Chapter 1 – Project Overview

This document describes the design, development, and validation of the Bank Loan Analytics Dashboard, an end-to-end Business Intelligence (BI) solution built for a fictional retail lender referred to as INEX Bank. The solution takes raw loan-level transaction data, validates key business metrics using SQL Server, and presents the results through an interactive three-page Power BI report.

The report is organized into three pages — Summary, Overview, and Details — each serving a different audience and purpose, ranging from executive KPI monitoring to transaction-level investigation. This mirrors how BI solutions are typically structured in real lending and financial-services organizations.

This documentation is written as a project record suitable for a professional portfolio and reflects a skill level consistent with roughly one year of hands-on experience in SQL and Power BI development.

Chapter 2 – Objective & Scope

The primary objective of the project is to convert raw, unstructured loan application data into a decision-ready reporting solution that allows management, credit-risk, and operations teams to:

- Track lending volume and disbursement performance.
- Monitor loan repayment and portfolio health (Good Loan vs Bad Loan).
- Compare current performance against the prior month using Time Intelligence (MTD, PMTD, MoM).
- Analyze lending patterns by geography, employment length, home ownership, and loan purpose.
- Drill down to individual loan records for audit and validation purposes.

The scope of the project covers data validation in SQL Server, data preparation in Power Query, data modeling and measure development in DAX, and dashboard design in Power BI Desktop. It does not include automated data refresh, row-level security, or deployment to Power BI Service — these are listed as future enhancements in Chapter 14.

Chapter 3 – Tools & Technologies Used

Tool	Purpose
SQL Server / SSMS	Source data storage and independent validation of business metrics
Power Query	Data cleaning, shaping, and loading into the Power BI data model
Power BI Desktop	Data modeling, DAX measures, and dashboard/report design
DAX	Calculated measures, KPIs, and Time Intelligence (MTD, PMTD, MoM)
Date (Calendar) Table	Supports Time Intelligence and consistent date-based filtering

Part 2 — Data Understanding & SQL Validation

Chapter 4 – Dataset Description

The dataset used for this project is a single table, `Bank_Loan_data`, containing one row per loan application. The key fields referenced throughout the analysis are summarized below.

Field	Description
id	Unique loan application identifier
issue_date	Date the loan was issued/approved
loan_amount	Amount funded (disbursed) to the borrower
total_payment	Total amount repaid by the borrower to date
int_rate	Interest rate applied to the loan (stored as a decimal, e.g. 0.12 = 12%)
dti	Debt-to-Income ratio of the borrower (stored as a decimal)
loan_status	Current status: Fully Paid, Current, or Charged Off
term	Loan repayment term: 36 months or 60 months
grade / sub_grade	Credit grade assigned to the loan (A–G, with sub-grades 1–5)
emp_length	Borrower's length of employment
home_ownership	Borrower's home ownership status (RENT, OWN, MORTGAGE)
purpose	Stated reason for the loan (debt consolidation, credit card, etc.)
address_state	U.S. state of the borrower

Chapter 5 – SQL-Based Data Validation

Before any metric was built in Power BI, it was first calculated directly in SQL Server against the raw table. This is standard practice in BI development: it establishes an independent, trusted baseline so that DAX measures can be checked against a known-correct value rather than trusted blindly. The full validation script is organized below by business metric.

5.1 Loan Volume

```
SELECT COUNT(id) AS Total_Loan_Application
FROM Bank_Loan_data;

SELECT COUNT(id) AS PMTD_Total_Loan_Application
FROM Bank_Loan_data
WHERE MONTH(issue_date) = 11 AND YEAR(issue_date) = 2021;
```

Note: The uploaded script filters the PMTD (Previous Month-to-Date) loan-application count using *MONTH = 12*, which is the same month used for the current MTD figure. Since the reporting month is December 2021, the previous month is November, so this filter should read *MONTH(issue_date) = 11 AND YEAR(issue_date) = 2021*, as corrected above.

5.2 Funded Amount

```
SELECT SUM(loan_amount) AS MTD_Total_Funded_Amount
FROM Bank_Loan_data
WHERE MONTH(issue_date) = 12 AND YEAR(issue_date) = 2021;

SELECT SUM(loan_amount) AS PMTD_Total_Funded_Amount
FROM Bank_Loan_data
WHERE MONTH(issue_date) = 11 AND YEAR(issue_date) = 2021;
```

These two queries are correct as written and form the basis of the MTD / PMTD funded-amount comparison shown on every dashboard page.

5.3 Amount Received

```
SELECT SUM(total_payment) AS Total_Amount_Received
FROM Bank_Loan_data;

SELECT SUM(total_payment) AS MTD_Total_Amount_Received
FROM Bank_Loan_data
WHERE MONTH(issue_date) = 12 AND YEAR(issue_date) = 2021;

SELECT SUM(total_payment) AS PMTD_Total_Amount_Received
```

```
FROM Bank_Loan_data
WHERE MONTH(issue_date) = 11 AND YEAR(issue_date) = 2021;
```

Note: As with Section 5.1, the original PMTD query for Amount Received also filtered on MONTH = 12 instead of MONTH = 11. This has been corrected above so the PMTD figure genuinely reflects November 2021, and matches the pattern used for Funded Amount.

5.4 Average Interest Rate & Average DTI

```
SELECT ROUND(AVG(int_rate), 4) * 100 AS Avg_Interest FROM Bank_Loan_data;

SELECT ROUND(AVG(int_rate), 4) * 100 AS MTD_Avg_Interest
FROM Bank_Loan_data WHERE MONTH(issue_date) = 12 AND YEAR(issue_date) = 2021;

SELECT ROUND(AVG(int_rate), 4) * 100 AS PMTD_Avg_Interest
FROM Bank_Loan_data WHERE MONTH(issue_date) = 11 AND YEAR(issue_date) = 2021;

SELECT ROUND(AVG(dti), 4) * 100 AS Avg_DTI FROM Bank_Loan_data;

SELECT ROUND(AVG(dti), 4) * 100 AS MTD_Avg_DTI
FROM Bank_Loan_data WHERE MONTH(issue_date) = 12 AND YEAR(issue_date) = 2021;

SELECT ROUND(AVG(dti), 4) * 100 AS PMTD_Avg_DTI
FROM Bank_Loan_data WHERE MONTH(issue_date) = 11 AND YEAR(issue_date) = 2021;
```

Because int_rate and dti are stored as decimals (e.g. 0.12), multiplying the rounded average by 100 correctly converts the value into a readable percentage.

5.5 Good Loan vs Bad Loan

"Good" loans are defined as loans with a status of Fully Paid or Current (i.e. not in default), while "Bad" loans are those that have been Charged Off.

```
SELECT (COUNT(CASE WHEN loan_status IN ('Fully Paid','Current') THEN id END) * 100.0)
      / COUNT(id) AS Good_Loan_Percent
FROM Bank_Loan_data;

SELECT COUNT(id) AS Good_Loan_Application
FROM Bank_Loan_data WHERE loan_status IN ('Fully Paid','Current');

SELECT SUM(loan_amount) AS Good_Loan_Funded_Amount
FROM Bank_Loan_data WHERE loan_status IN ('Fully Paid','Current');

SELECT SUM(total_payment) AS Good_Loan_Received_Amount
FROM Bank_Loan_data WHERE loan_status IN ('Fully Paid','Current');

SELECT (COUNT(CASE WHEN loan_status = 'Charged Off' THEN id END) * 100.0)
      / COUNT(id) AS Bad_Loan_Percent
FROM Bank_Loan_data;

SELECT COUNT(id) AS Bad_Loan_Application
FROM Bank_Loan_data WHERE loan_status = 'Charged Off';

SELECT SUM(loan_amount) AS Bad_Loan_Funded_Amount
FROM Bank_Loan_data WHERE loan_status = 'Charged Off';

SELECT SUM(total_payment) AS Bad_Loan_Received_Amount
FROM Bank_Loan_data WHERE loan_status = 'Charged Off';
```

Note: The original Good_Loan_Percent query used integer division (COUNT(...) * 100) without a decimal literal, which risks truncating the result to a whole number on some SQL Server configurations. Using

100.0, as in the Bad_Loan_Percent query and shown above, forces decimal division and returns an accurate percentage — this was applied consistently in the corrected version above.

5.6 Breakdown Queries

A set of GROUP BY queries was used to validate every categorical breakdown shown on the Overview and Summary dashboards: loan status, month, address state, term, employee length, purpose, and home ownership. As an example, the loan status breakdown query is shown below; the remaining breakdowns follow the same pattern with the relevant grouping column.

```
SELECT loan_status,
       COUNT(id)           AS Total_Loan_Applications,
       SUM(total_payment)  AS Total_Amount_Received,
       SUM(loan_amount)    AS Total_Funded_Amount,
       AVG(int_rate * 100) AS Interest_Rate,
       AVG(dti * 100)      AS DTI
FROM Bank_Loan_data
GROUP BY loan_status;
```

This query result was checked directly against the Loan Status table on the Summary Dashboard (Chapter 8) and the two figures matched, confirming that the Power BI measures were built correctly.

Minor naming inconsistencies were also present in the original script (e.g. "total_amount_recieve", "Total_Amount_Received") — these are simple aliasing typos in SQL and do not affect the query logic, but have been spelled correctly throughout this document and are worth cleaning up in the source .sql file for readability.

Part 3 — Data Modeling & DAX

Chapter 6 – Data Model

The data model consists of the Bank_Loan_data fact table and a dedicated Date Table, joined through a one-to-many relationship on the issue date. A separate Date table (rather than relying on issue_date directly) is standard Power BI practice because it:

- Provides a complete, continuous calendar even where the fact table has gaps.
- Enables built-in and custom Time Intelligence functions such as MTD, PMTD, and MoM.
- Allows filtering and slicing consistently across every visual and page in the report.

The relationship was set as one-to-many (Date Table to Bank_Loan_data) with a single filter direction, which is the correct configuration for this type of star-schema model.

Chapter 7 – DAX Measures & Time Intelligence

Reusable DAX measures were created for each KPI shown on the dashboards, rather than relying on implicit measures. This makes the report easier to maintain and reuse across visuals. The core measures include:

- Total Loan Applications = COUNT(Bank_Loan_data[id])
- Total Funded Amount = SUM(Bank_Loan_data[loan_amount])
- Total Amount Received = SUM(Bank_Loan_data[total_payment])
- Average Interest Rate = AVERAGE(Bank_Loan_data[int_rate])
- Average DTI = AVERAGE(Bank_Loan_data[dti])

For each of these base measures, MTD and PMTD variants were created using Time Intelligence functions (such as TOTALMTD and DATEADD) against the Date Table, and MoM (Month-over-Month) variance measures were derived by comparing MTD to PMTD. Good Loan and Bad Loan measures were built using CALCULATE with a filter on loan_status, mirroring the logic validated in the SQL queries in Chapter 5.

A Field Parameter was also used on the "Select Measure" control (visible on the left of every page) so that users can switch the underlying measure — for example, toggling between Total Funded Amount and Total Amount Received — without needing a separate visual for each option.

Part 4 — Summary Dashboard

Chapter 8 – Summary Dashboard

The Summary Dashboard is the executive landing page of the report. It is designed to answer, at a glance, "How is the loan portfolio performing overall, and how healthy is it?"

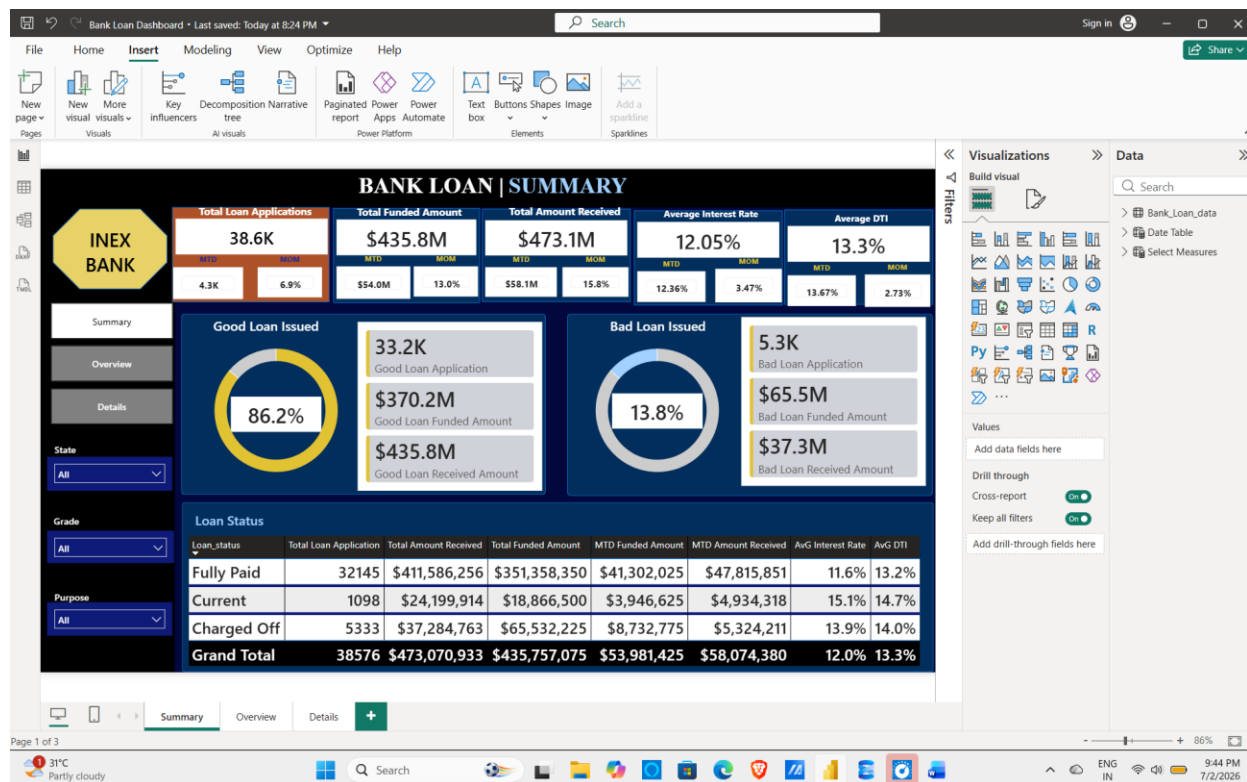


Figure 8.1 – Summary Dashboard

8.1 KPI Cards

Across the top of the page, five KPI cards display the headline numbers for the full portfolio, each paired with its MTD (Month-to-Date) and MoM (Month-over-Month) figures:

KPI	Total	MTD	MoM
Total Loan Applications	38.6K	4.3K	6.9%
Total Funded Amount	\$435.8M	\$54.0M	13.0%
Total Amount Received	\$473.1M	\$58.1M	15.8%
Average Interest Rate	12.05%	12.36%	3.47%
Average DTI	13.3%	13.67%	2.73%

A useful sense-check: Total Amount Received (\$473.1M) exceeds Total Funded Amount (\$435.8M), which is expected — repayments include interest on top of principal, so the received figure should be higher than the amount originally disbursed.

8.2 Good Loan vs Bad Loan

Two donut charts summarize portfolio quality:

- Good Loan Issued — 86.2% of loans, 33.2K applications, \$370.2M funded, \$435.8M received.
- Bad Loan Issued — 13.8% of loans, 5.3K applications, \$65.5M funded, only \$37.3M received.

This is the single most important risk indicator on the page: for Bad (Charged Off) loans, the bank recovered noticeably less (\$37.3M) than it originally disbursed (\$65.5M), representing a realized loss on that segment. For Good loans, the opposite is true — received amount exceeds funded amount, reflecting normal interest income on performing loans.

8.3 Loan Status Table

A detailed breakdown table lists every loan_status category with its application count, received amount, funded amount, MTD figures, average interest rate, and average DTI, rolling up to a Grand Total row. This table was cross-checked against the SQL query in Section 5.6 and the totals reconcile exactly (38,576 applications and \$435.8M / \$473.1M in the Grand Total row).

Part 5 — Overview Dashboard

Chapter 9 – Overview Dashboard

The Overview Dashboard is the analytical page of the report. Where the Summary page answers "what happened," the Overview page helps answer "where and why." The screenshot below has the Good Vs Bad Loan slicer set to Good Loan, which is why the KPI cards on this page (33.2K applications, \$370.2M funded) match the Good Loan figures from the Summary page.

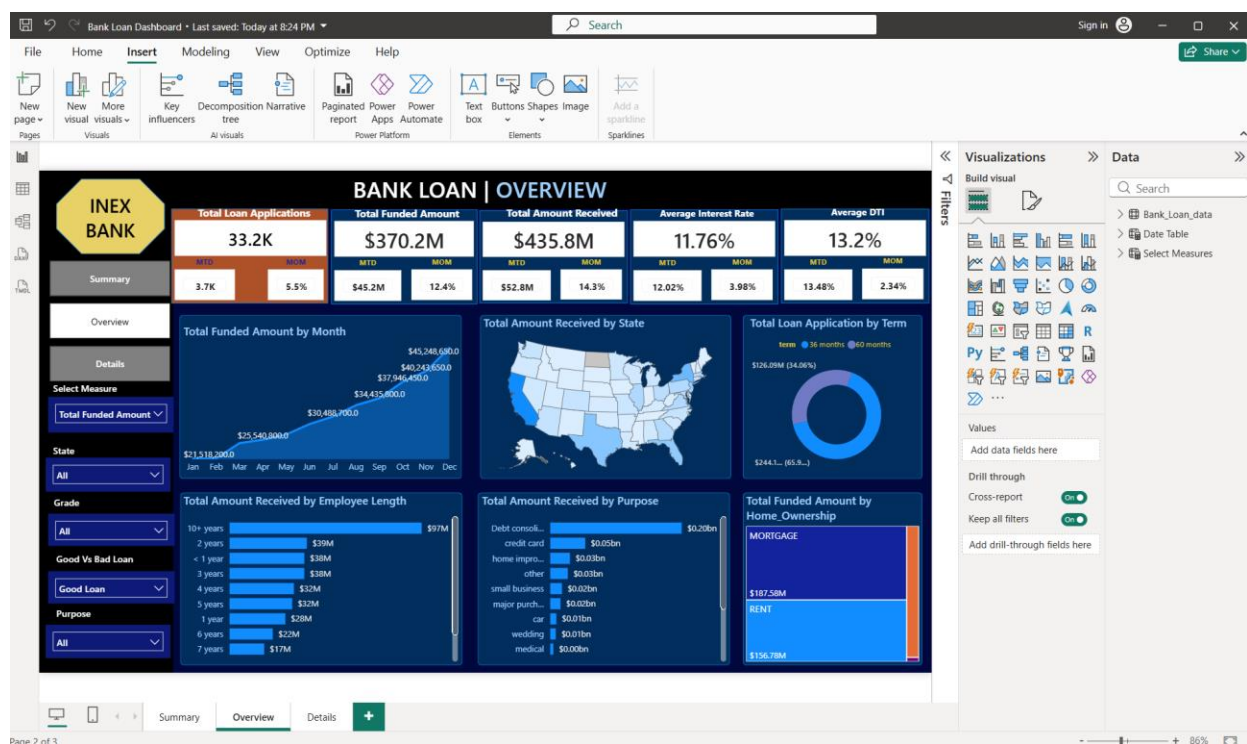


Figure 9.1 – Overview Dashboard (filtered to Good Loan)

9.1 Visuals on this Page

Visual	Insight it Provides
Total Funded Amount by Month (line chart)	Shows the monthly growth trend in lending, rising from ~\$21.5M in January to ~\$45.2M by December
Total Amount Received by State (shape map)	Highlights which U.S. states contribute the most in repayments, useful for regional performance tracking
Total Loan Application by Term (donut)	Splits applications between 36-month and 60-month terms
Total Amount Received by Employee Length (bar chart)	Shows that borrowers with 10+ years of employment history contribute the largest share of repayments (\$97M)
Total Amount Received by Purpose (bar chart)	Debt consolidation dominates the portfolio (~\$0.20bn), far ahead of credit card, home improvement, and other purposes
Total Funded Amount by Home Ownership (bar chart)	Mortgage holders (\$187.58M) and renters (\$156.78M) account for almost all funded amount

9.2 Left-Panel Filters

Every page shares the same left-hand filter panel — State, Grade, Good Vs Bad Loan, and Purpose — along with the Select Measure field parameter. Because these filters are placed outside the report canvas on every page and Power BI's default sync-filter behavior is used, a selection made here stays in effect as the user navigates between Summary, Overview, and Details, giving the report a consistent, application-like feel.

Part 6 — Details Dashboard, Insights & Conclusion

Chapter 10 – Details Dashboard

11.1 Dashboard Overview

The Details Dashboard serves as the operational reporting layer of the Bank Loan Analytics solution. While the Summary Dashboard focuses on executive KPIs and the Overview Dashboard provides analytical insights, the Details Dashboard allows users to access transaction-level information for every loan record.

This dashboard enables business users to drill down into individual loan applications, verify aggregated metrics, and investigate specific customer records without navigating away from the reporting environment.

Unlike visual-heavy dashboards, the Details Dashboard prioritizes data accuracy, transparency, and accessibility through a structured tabular layout.

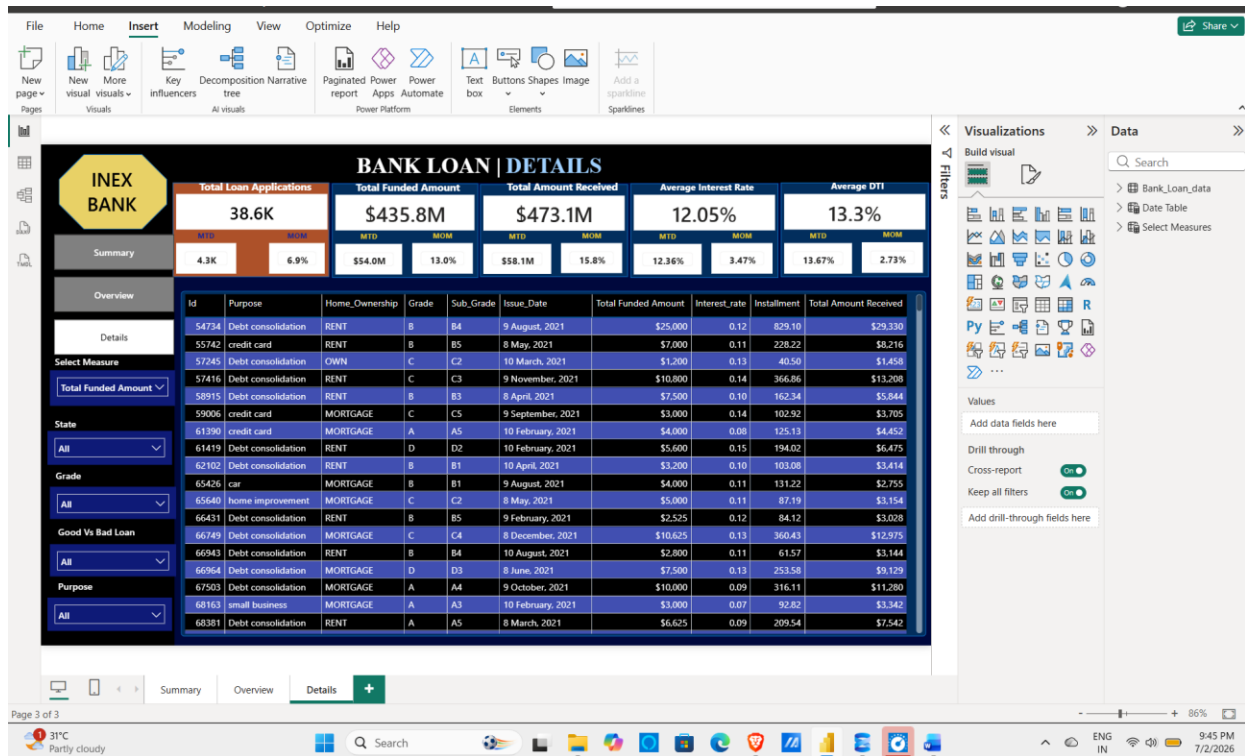


Figure 11.1 – Details Dashboard

11.2 Business Objective

The primary objective of the Details Dashboard is to provide a centralized view of individual loan records, allowing users to:

- Review complete loan information.
- Validate summarized dashboard metrics.
- Perform operational analysis.
- Investigate customer-specific transactions.
- Export data for further reporting if required.

By combining transaction-level visibility with interactive filtering, the dashboard supports both operational teams and management.

11.3 Report Layout

The dashboard consists of a detailed table visual containing key attributes from the loan dataset:

Column	Description
Loan ID	Unique identifier of each loan application
Purpose	Reason for the loan application
Home Ownership	Customer's home ownership category
Grade	Credit grade assigned to the loan
Sub Grade	Detailed credit classification
Issue Date	Loan approval date
Funded Amount	Amount sanctioned by the bank
Interest Rate	Interest charged on the loan
Installment	Monthly repayment amount
Total Amount Received	Total repayment collected from the customer

The table updates dynamically based on slicer selections and interactions from other dashboard pages.

11.4 Why a Table Visual?

Although charts are effective for summarizing information, they cannot display detailed transaction-level records. A table visual was selected because it presents complete loan information in a structured format, supports operational validation, enables detailed comparisons between individual records, improves report transparency, and complements the executive and analytical dashboards. This combination of summary, analytical, and detailed reporting provides a complete Business Intelligence solution.

11.5 Interactive Navigation

To improve usability, page navigator buttons were implemented, allowing users to switch between the Summary, Overview, and Details Dashboards without relying on Power BI's default page tabs. This offers a more professional user experience, consistent navigation, faster access to reports, a cleaner dashboard layout, and improved presentation quality — transforming the report into a complete BI application rather than a collection of independent pages.

11.6 Company Branding

To improve presentation quality and maintain a professional appearance, a bank logo was added to the report. Branding provides a professional report identity, improved visual consistency, and better stakeholder presentation. Although branding does not affect analytics, it contributes to the overall usability and professionalism of the solution.

11.7 Dashboard Interactivity

The Details Dashboard remains fully synchronized with filters applied across the report. Users can select a specific state, filter by loan grade, choose a loan purpose, or navigate from the Summary or Overview pages — all selected filters automatically update the table, ensuring users view only the relevant records. This behavior is made possible through Power BI's filter propagation and page-level interaction capabilities.

11.8 Business Value

The Details Dashboard supports operational decision-making by providing complete transparency into the underlying data. Business users can validate KPI calculations, investigate specific customer records, analyze loan characteristics, review repayment details, and verify dashboard accuracy — bridging the gap between high-level analytics and raw transactional data.

11.9 Chapter Summary

The Details Dashboard provides transaction-level visibility into the bank's loan portfolio. By combining structured tabular reporting with interactive navigation and synchronized filtering, it enables users to investigate individual loan records while maintaining consistency with the executive and analytical dashboards.

Chapter 11 – Business Insights

12.1 Overview

A Business Intelligence dashboard is valuable not only for presenting data but also for enabling actionable insights. Through interactive reporting and KPI monitoring, the Bank Loan Analytics Dashboard helps stakeholders understand lending behavior, evaluate portfolio performance, and identify areas for improvement. The following observations illustrate the analytical capabilities of the solution and should be interpreted based on the data available at the time of analysis.

12.2 Lending Performance

The dashboard enables management to monitor total loan applications received, total capital disbursed, and customer repayment performance. By comparing Month-to-Date (MTD) and Previous Month-to-Date (PMTD) metrics, stakeholders can quickly identify growth trends or declines in lending activity.

12.3 Portfolio Quality

The Good Loan vs. Bad Loan analysis provides a clear picture of portfolio health, measuring the proportion of performing loans, monitoring non-performing loans, assessing overall lending risk, and supporting credit policy evaluation. A higher percentage of Good Loans generally indicates a healthier portfolio and stronger repayment performance — in this dataset, 86.2% of loans are performing.

12.4 Geographic Insights

The shape map highlights variations in lending activity across different states, enabling management to identify high-performing regions, detect underperforming markets, allocate marketing efforts strategically, and support regional expansion planning.

12.5 Customer Segmentation

Employment Length, Home Ownership, and Loan Purpose visualizations provide valuable insight into customer characteristics, helping answer questions such as which customer segments apply for the highest number of loans, which employment groups receive larger funded amounts, which loan purposes dominate the portfolio, and how housing status relates to lending activity. Such insights can guide customer acquisition strategy and product development.

12.6 Financial Performance

The dashboard enables continuous monitoring of Total Funded Amount, Total Amount Received, Average Interest Rate, and Average DTI — KPIs that support profitability analysis, repayment monitoring, and risk assessment.

12.7 Time-Based Analysis

Using Time Intelligence calculations, stakeholders can evaluate monthly growth, seasonal lending patterns, repayment trends, and changes in customer demand. Comparing MTD, PMTD, and MoM metrics provides a comprehensive view of business performance over time.

12.8 Business Impact

The implemented solution delivers centralized reporting, reduced manual reporting effort, improved decision-making, enhanced data transparency, interactive self-service analytics, and faster access to business insights — enabling stakeholders to transition from static reporting to dynamic, data-driven decision-making.

Chapter 12 – Challenges & Learnings

13.1 Challenges Faced

- Data Validation — ensuring that Power BI calculations matched SQL Server query results required careful validation of business metrics (see Chapter 5, including two PMTD queries that needed correcting).
- Time Intelligence — implementing MTD, PMTD, and MoM calculations required a dedicated Date Table and correctly configured relationships.

- Data Modeling — establishing the appropriate one-to-many relationship between the Date Table and loan data was essential for accurate filtering and Time Intelligence functions.
- Dashboard Design — balancing information density with visual clarity required thoughtful layout planning, consistent formatting, and appropriate chart selection.
- Dynamic Reporting — implementing field parameters and synchronized visual interactions increased report flexibility while maintaining usability.

13.2 Key Learnings

This project strengthened practical knowledge in Business Intelligence concepts, SQL-based data validation, Power Query transformations, data modeling, DAX programming, Time Intelligence, dashboard design principles, interactive reporting, and business storytelling through data. Beyond technical skills, the project emphasized the importance of understanding business requirements before designing analytical solutions.

Chapter 13 – Future Enhancements

Although the current dashboard provides comprehensive reporting capabilities, several enhancements could further improve the solution:

- Incremental data refresh.
- Row-Level Security (RLS).
- Drill-through reports.
- Tooltips with additional metrics.
- Forecasting using Power BI Analytics.
- Integration with Azure SQL Database.
- Automated data refresh through Power BI Service.
- Mobile-optimized report layout.
- Predictive analytics using Azure Machine Learning.
- AI-powered anomaly detection.

These enhancements would improve scalability, security, and analytical capabilities for enterprise deployment.

Chapter 14 – Conclusion

The Bank Loan Analytics Dashboard successfully demonstrates the development of an end-to-end Business Intelligence solution using SQL Server, Power Query, DAX, and Power BI. Beginning with SQL-based data validation and progressing through data transformation, modeling, KPI development, and dashboard design, the project showcases the complete BI development lifecycle.

The resulting solution provides executive reporting, interactive analytical dashboards, and transaction-level visibility, enabling stakeholders to monitor lending performance, assess portfolio quality, and make informed business decisions. By implementing reusable DAX measures, Time Intelligence calculations, dynamic field parameters, interactive visualizations, and consistent design principles, the project delivers a scalable and maintainable reporting solution that aligns with industry best practices.

This project not only strengthened technical expertise in Power BI and SQL Server but also reinforced the importance of translating business requirements into meaningful analytical solutions. It serves as a

practical demonstration of Business Intelligence concepts and reflects the skills required to develop professional reporting solutions in a real-world environment.

Appendix A – Consolidated & Corrected SQL Script

The queries below consolidate the full validation script from Chapter 5 into a single, corrected reference file — with the two PMTD filters and the integer-division issue fixed, and column aliases spelled correctly.

```
-- Loan volume
SELECT COUNT(id) AS Total_Loan_Application FROM Bank_Loan_data;

SELECT COUNT(id) AS PMTD_Total_Loan_Application FROM Bank_Loan_data
WHERE MONTH(issue_date) = 11 AND YEAR(issue_date) = 2021;  -- corrected: was 12

-- Funded amount
SELECT SUM(loan_amount) AS MTD_Total_Funded_Amount FROM Bank_Loan_data
WHERE MONTH(issue_date) = 12 AND YEAR(issue_date) = 2021;

SELECT SUM(loan_amount) AS PMTD_Total_Funded_Amount FROM Bank_Loan_data
WHERE MONTH(issue_date) = 11 AND YEAR(issue_date) = 2021;

-- Amount received
SELECT SUM(total_payment) AS Total_Amount_Received FROM Bank_Loan_data;

SELECT SUM(total_payment) AS MTD_Total_Amount_Received FROM Bank_Loan_data
WHERE MONTH(issue_date) = 12 AND YEAR(issue_date) = 2021;

SELECT SUM(total_payment) AS PMTD_Total_Amount_Received FROM Bank_Loan_data
WHERE MONTH(issue_date) = 11 AND YEAR(issue_date) = 2021;  -- corrected: was 12

-- Average interest rate & DTI
SELECT ROUND(AVG(int_rate), 4) * 100 AS Avg_Interest FROM Bank_Loan_data;
SELECT ROUND(AVG(int_rate), 4) * 100 AS MTD_Avg_Interest FROM Bank_Loan_data
WHERE MONTH(issue_date) = 12 AND YEAR(issue_date) = 2021;
SELECT ROUND(AVG(int_rate), 4) * 100 AS PMTD_Avg_Interest FROM Bank_Loan_data
WHERE MONTH(issue_date) = 11 AND YEAR(issue_date) = 2021;

SELECT ROUND(AVG(dti), 4) * 100 AS Avg_DTI FROM Bank_Loan_data;
SELECT ROUND(AVG(dti), 4) * 100 AS MTD_Avg_DTI FROM Bank_Loan_data
WHERE MONTH(issue_date) = 12 AND YEAR(issue_date) = 2021;
SELECT ROUND(AVG(dti), 4) * 100 AS PMTD_Avg_DTI FROM Bank_Loan_data
WHERE MONTH(issue_date) = 11 AND YEAR(issue_date) = 2021;

-- Good loan vs bad loan
SELECT (COUNT(CASE WHEN loan_status IN ('Fully Paid','Current') THEN id END) * 100.0)
      / COUNT(id) AS Good_Loan_Percent FROM Bank_Loan_data;  -- corrected: *100.0

SELECT COUNT(id) AS Good_Loan_Application FROM Bank_Loan_data
WHERE loan_status IN ('Fully Paid','Current');

SELECT SUM(loan_amount) AS Good_Loan_Funded_Amount FROM Bank_Loan_data
WHERE loan_status IN ('Fully Paid','Current');

SELECT SUM(total_payment) AS Good_Loan_Received_Amount FROM Bank_Loan_data
WHERE loan_status IN ('Fully Paid','Current');

SELECT (COUNT(CASE WHEN loan_status = 'Charged Off' THEN id END) * 100.0)
      / COUNT(id) AS Bad_Loan_Percent FROM Bank_Loan_data;

SELECT COUNT(id) AS Bad_Loan_Application FROM Bank_Loan_data
WHERE loan_status = 'Charged Off';

SELECT SUM(loan_amount) AS Bad_Loan_Funded_Amount FROM Bank_Loan_data
WHERE loan_status = 'Charged Off';
```



```

SELECT SUM(total_payment) AS Bad_Loan_Received_Amount FROM Bank_Loan_data
WHERE loan_status = 'Charged Off';

-- Breakdown queries
SELECT loan_status, COUNT(id) AS Total_Loan_Applications,
       SUM(total_payment) AS Total_Amount_Received,
       SUM(loan_amount) AS Total_Funded_Amount,
       AVG(int_rate * 100) AS Interest_Rate, AVG(dti * 100) AS DTI
FROM Bank_Loan_data GROUP BY loan_status;

SELECT MONTH(issue_date) AS Month_Num, DATENAME(MONTH, issue_date) AS Month_Name,
       COUNT(id) AS Total_Loan_Application, SUM(loan_amount) AS Total_Funded_Amount,
       SUM(total_payment) AS Total_Amount_Received
FROM Bank_Loan_data GROUP BY MONTH(issue_date), DATENAME(MONTH, issue_date)
ORDER BY MONTH(issue_date);

SELECT address_state, COUNT(id) AS Total_Loan_Application,
       SUM(loan_amount) AS Total_Funded_Amount, SUM(total_payment) AS Total_Amount_Received
FROM Bank_Loan_data GROUP BY address_state ORDER BY COUNT(id) DESC;

SELECT term, COUNT(id) AS Total_Loan_Application,
       SUM(loan_amount) AS Total_Funded_Amount, SUM(total_payment) AS Total_Amount_Received
FROM Bank_Loan_data GROUP BY term ORDER BY term;

SELECT emp_length, COUNT(id) AS Total_Loan_Application,
       SUM(loan_amount) AS Total_Funded_Amount, SUM(total_payment) AS Total_Amount_Received
FROM Bank_Loan_data GROUP BY emp_length ORDER BY COUNT(id) DESC;

SELECT purpose, COUNT(id) AS Total_Loan_Application,
       SUM(loan_amount) AS Total_Funded_Amount, SUM(total_payment) AS Total_Amount_Received
FROM Bank_Loan_data GROUP BY purpose ORDER BY COUNT(id) DESC;

SELECT home_ownership, COUNT(id) AS Total_Loan_Application,
       SUM(loan_amount) AS Total_Funded_Amount, SUM(total_payment) AS Total_Amount_Received
FROM Bank_Loan_data GROUP BY home_ownership ORDER BY COUNT(id) DESC;

```